

The problem: Generic tools vs. specific business needs

Introduction

From built-in to custom capabilities

Your journey:

Part 1

Understood tools fundamentally
and built your first tool-enabled agent

Part 2

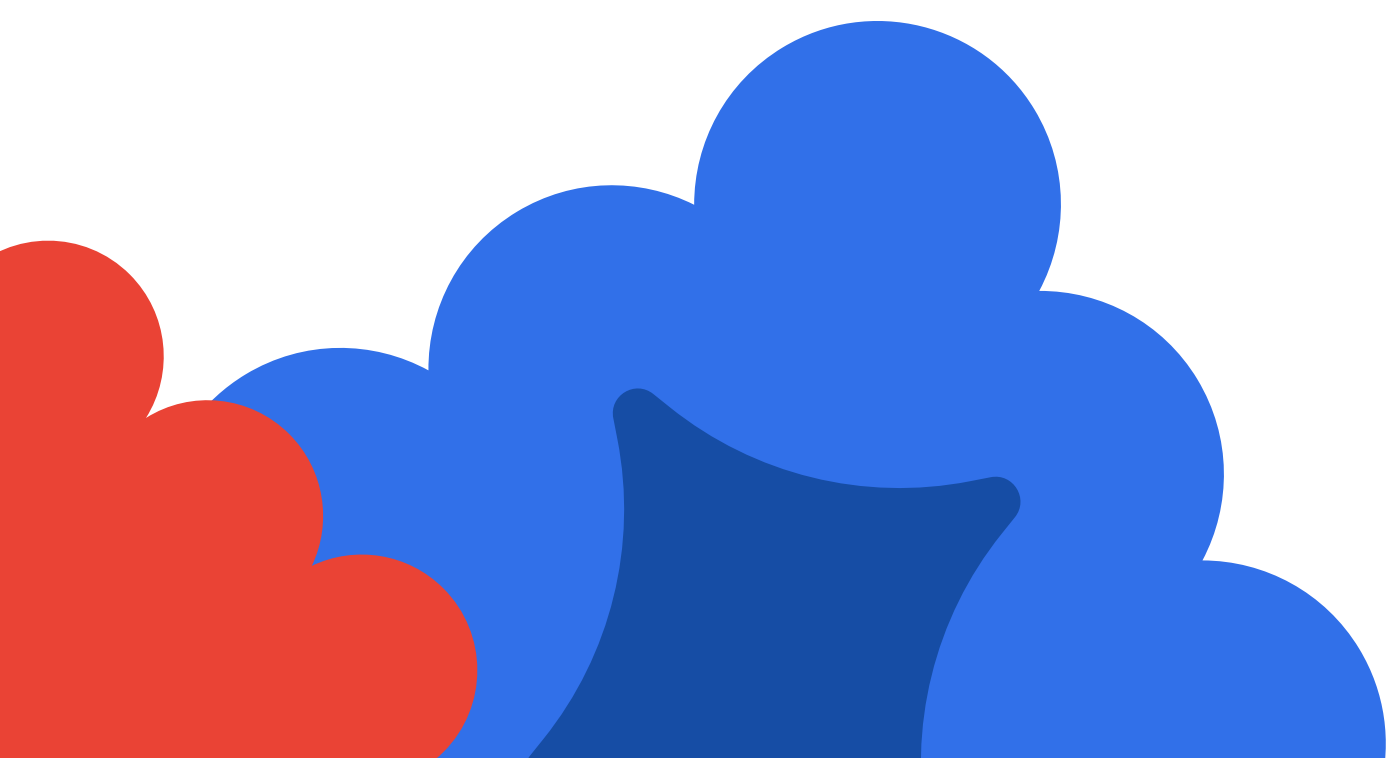
Used production-ready built-in tools
(Google Search, Code Execution)

Part 3

Create custom tools tailored to your
specific needs

The shift:

Instead of importing ready-made tools, you write
Python functions that ADK automatically converts
to tools.



The problem:

Generic tools vs. specific business needs

Built-in tools are excellent for common tasks:

```
Python
# Built-in tools handle universal needs
from google.adk.tools import google_search

agent = LlmAgent(
    model='gemini-2.5-flash',
    tools=[google_search] # Works for any web search
)
```

But what about YOUR specific business logic?

```
Python
# Your business needs unique tools:
# - Calculate shipping costs for YOUR products
# - Look up inventory in YOUR database
# - Process refunds in YOUR system
# - Validate customer loyalty points in YOUR program
# - Book appointments in YOUR calendar
# - Check order status in YOUR e-commerce platform

# Built-in tools can't know your business!
```

Issues with generic solutions:

- ✗ **No business context:** Built-in tools don't know your products, pricing, policies
- ✗ **Can't access your systems:** Your proprietary databases, APIs, internal services
- ✗ **One-size-fits-all:** Generic tools don't match your specific workflows
- ✗ **Limited customization:** Can't modify built-in tool behavior for your needs

The root problem

Every business has unique logic, data, and systems that require custom integration.

